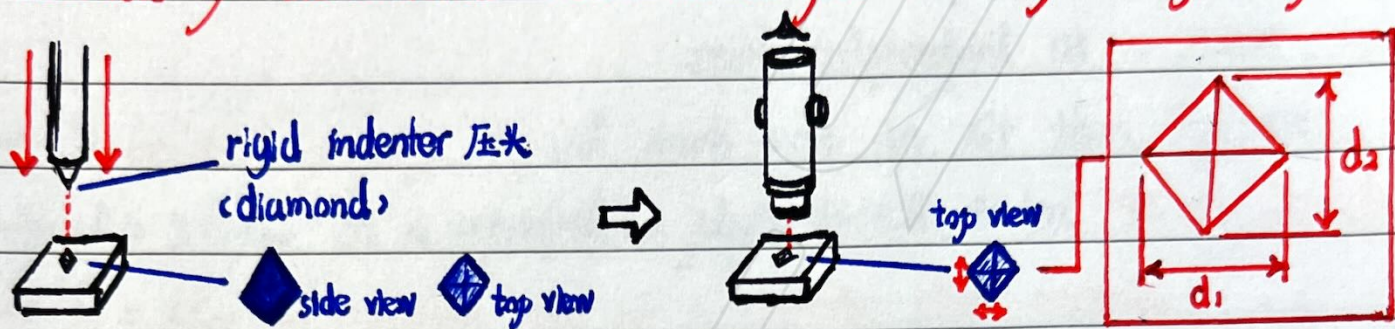


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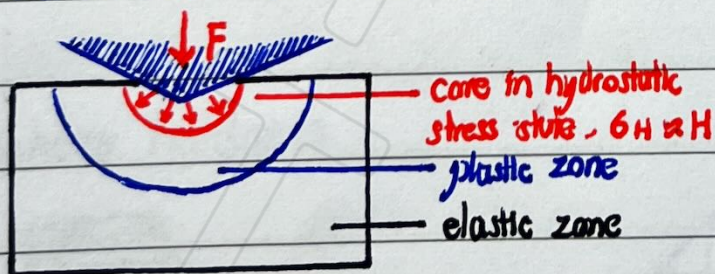
Measure the material hardness

• Hardness of the material = resistance to penetration by a rigid object



- Vickers square pyramid  / Rockwell cone  / Brinell sphere  / Berkovite triangular pyramid  / Knoop pyramid with rhomboidal base  / ... hardness tests
- hardness mm 10^{-3} m / micro-hardness μm 10^{-6} m / nano-hardness nm 10^{-9} m
→ use a Vickers micro-hardness test

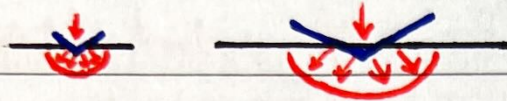
Yield stress with hardness



$$H = \frac{F}{A^*}$$

- * A = area of contact between indenter and material projected on the plane perpendicular to the applied load
→ base of the pyramidal indentation 压痕 footprint this case

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penetrate distance $\uparrow\uparrow\uparrow$



• shape of indentation footprint unchanged $\rightarrow$ self-similarity

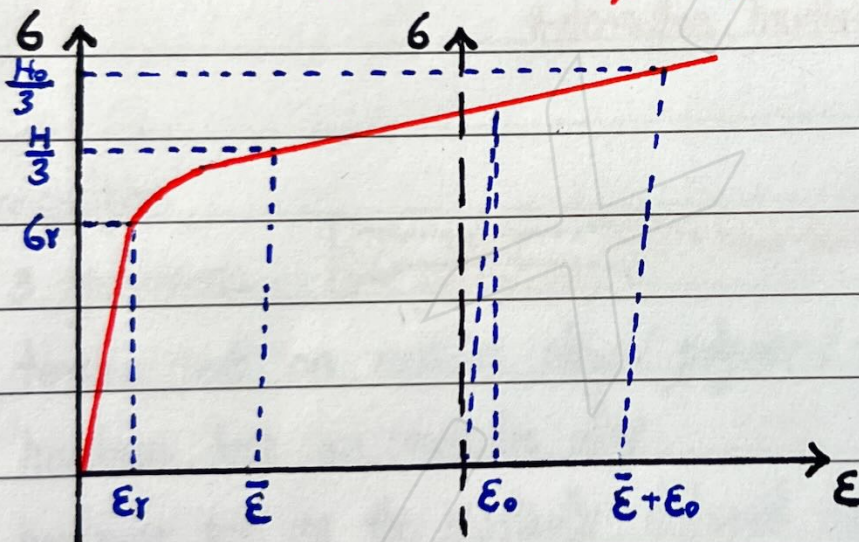
• $H \propto F$ or depth

scale independent as a stress

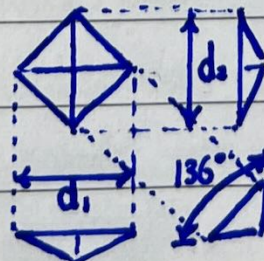
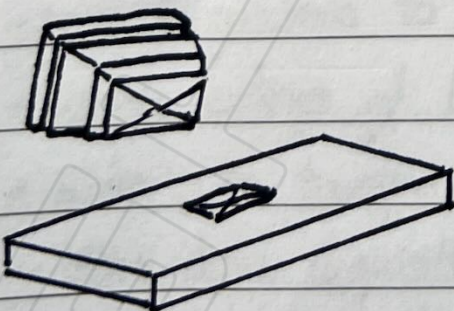
$\rightarrow$ H mechanical property depend much more on σ_r than stiffness

$\rightarrow H = 3\sigma_{flow}$ at a representative plastic strain $\bar{\epsilon}$ 8-10% 9% in this lab

$\rightarrow H = 3\sigma_r$ in particular case of a perfectly plastic material



Vickers hardness



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Vickers hardness HV = $\frac{\text{indentation Force}}{\text{entire area of contact}}$

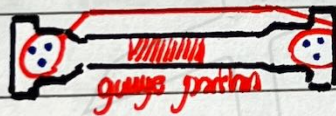
$$HV = \frac{F}{\frac{d^2}{2 \sin(\frac{1}{2} \times 136^\circ)}} = \frac{2 \sin(68^\circ) F}{d^2} \text{ (kgf/mm}^2\text{)}$$

xxx HV yy / zz
vickers hardness load time

* separate indentations to avoid the large plastic zones created by 6* footprint diagonal $\rightarrow$ 2mm between adjacent indentations in this lab

Procedure

- 3 specimens



- tension test on metallic alloy / polymer / composite
- hardness test on metallic alloy
- * hardness test on the plastically deformed metallic specimen / at different loads / on the polymeric material
- tension tests
 - connect specimen to the machine's jaws
 - balance readings of load and displacement gauges via software
 - * make sure transparent safety guards are in place
 - start data acquisition 获取 via software
 - slowly apply tensile load $\rightarrow$ V deformation = constant

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* $f_{max} = 1\text{Hz}$

- stop the data logger when specimen fails
- Remove the repeat with another specimen

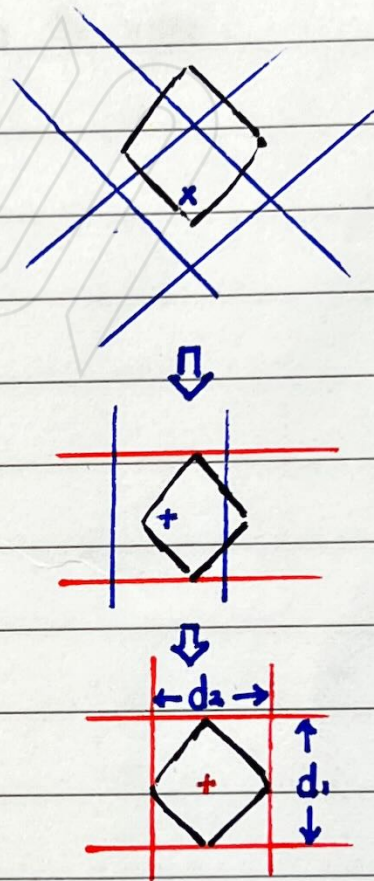
• Hardness tests

- polish 磨光 one side of the T-shaped ends with sand paper
- xxx HV1/10
- place specimen and focus the microscope, bring indenter above
- start test
- bring microscope above the indentation footprint when its over
- use software take a photograph then measure diagonals
- HV will be calculated automatically
- repeat for 3-5 times
- Graph 3 σ_t vs ϵ_t for tension test
- Graph 4/5/6 σ_n and σ_t vs ϵ_n and ϵ_t

11/28/2022 Tension and Hardness During Lab

Specimen	Number of attempt	d1 (μm)	d2 (μm)	d (μm)	HV (kgf/mm^2)	1st Width (mm)	2nd Width (mm)	3rd Width (mm)	Width (mm)	1st Height (mm)	2nd Height (mm)	3rd Height (mm)	Height (mm)	A (m^2)
Aluminum	1													
	2													
	3													
	4													
	5													
Carbon fiber														
Ploymer														

• adjust the machine by dial



* use different sand-paper with different items

+ do not think the surface is flat enough with the not-enough polishing, which might affect the results and cause the wide-spread range of the data

11/28/2022 Tension and Hardness During Lab

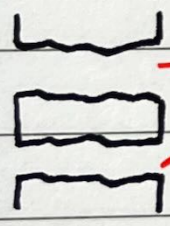
- wear goggles  before tests
- cannot observe the deform directly through the eyes

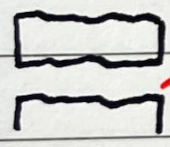
 looks like fluffs on the section

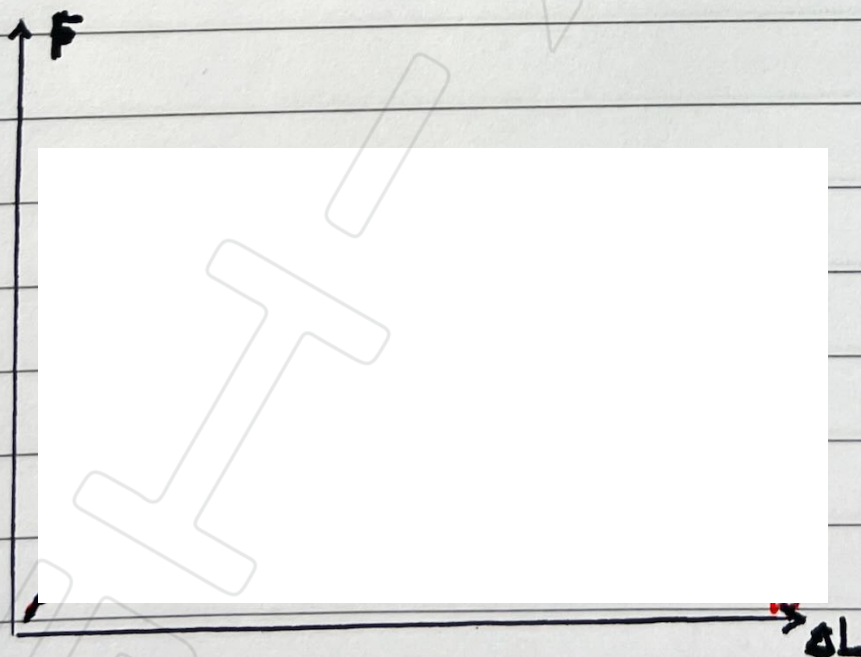
 carbon fibre when break

 smooth section

 aluminium alloy when break

 break into 3 pieces

 rigid body when break



- aluminium alloy
- carbon fibre
- rigid body
- not scaled,
- drawn independently

Reading displayed on the screen during the test

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7.00E+0
6.00E+0
5.00E+0
4.00E+0
3.00E+0
2.00E+0
1.00E+0
0.00E+0
0.

true stress σ (Pa)

true strain ϵ

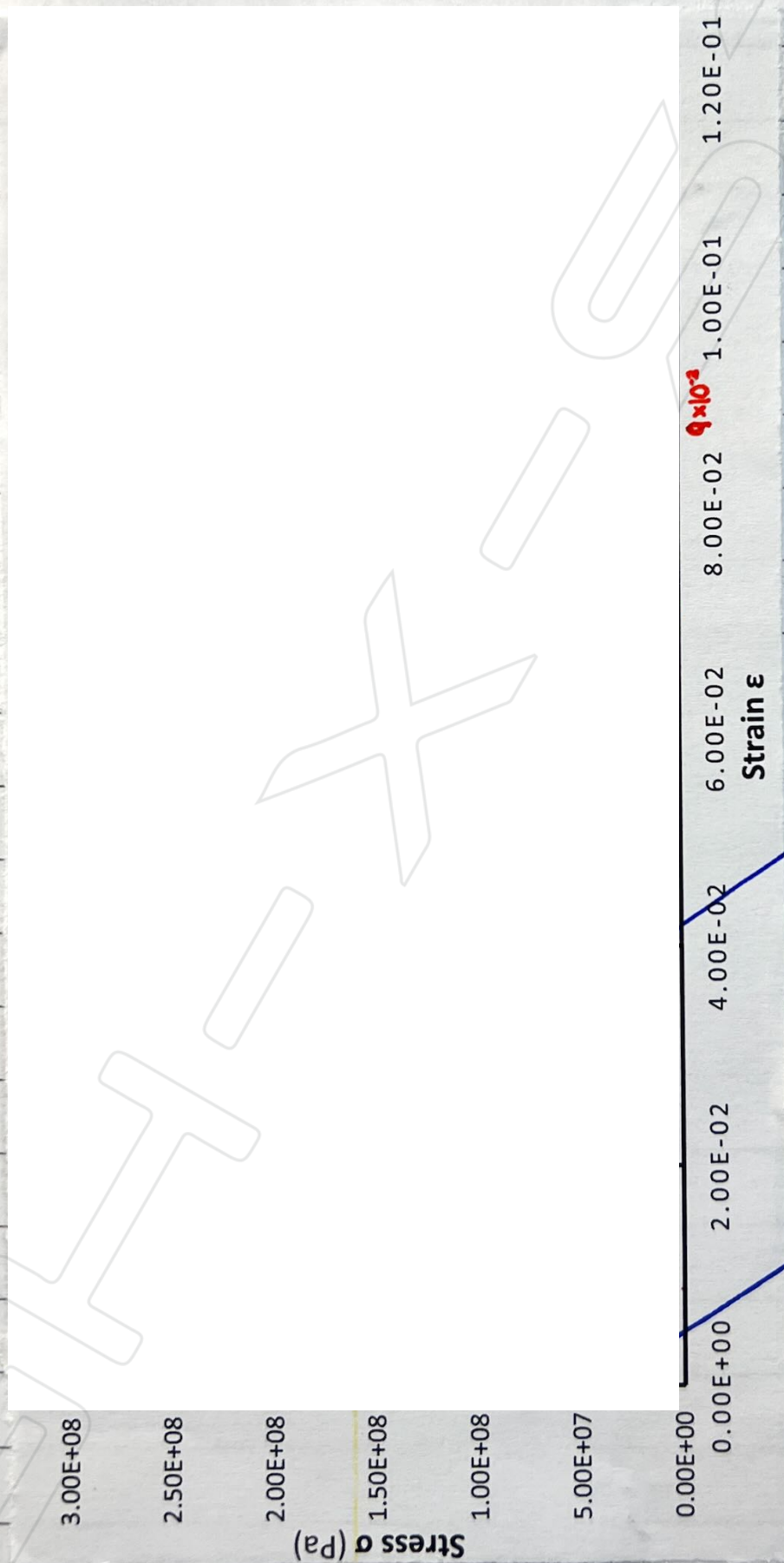
Al ductile
Carbon fiber
brittle
Polymer
ductile +

Stiffness
Carbon fibre
> Al >> Polymer

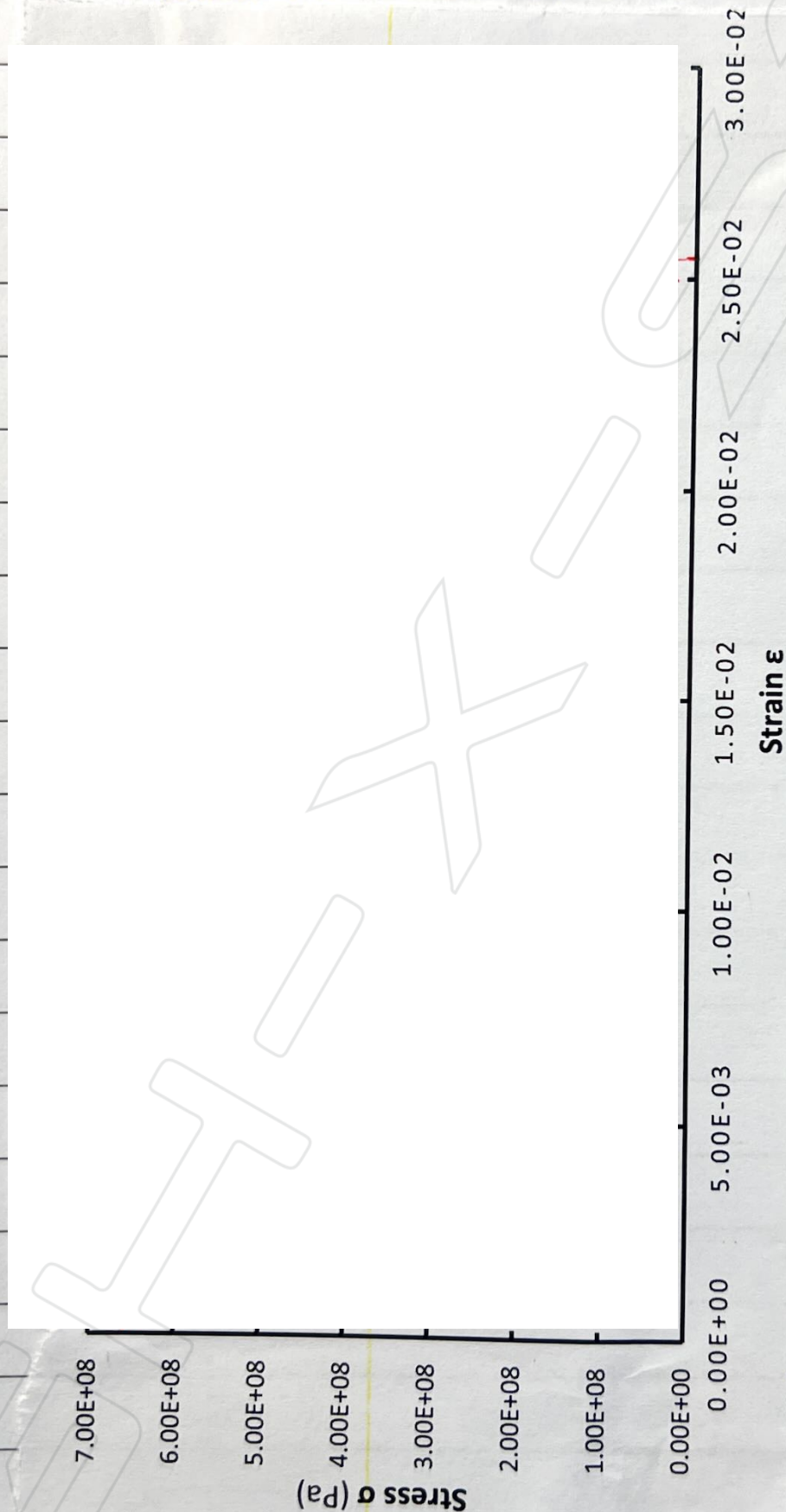
plastic

elastic

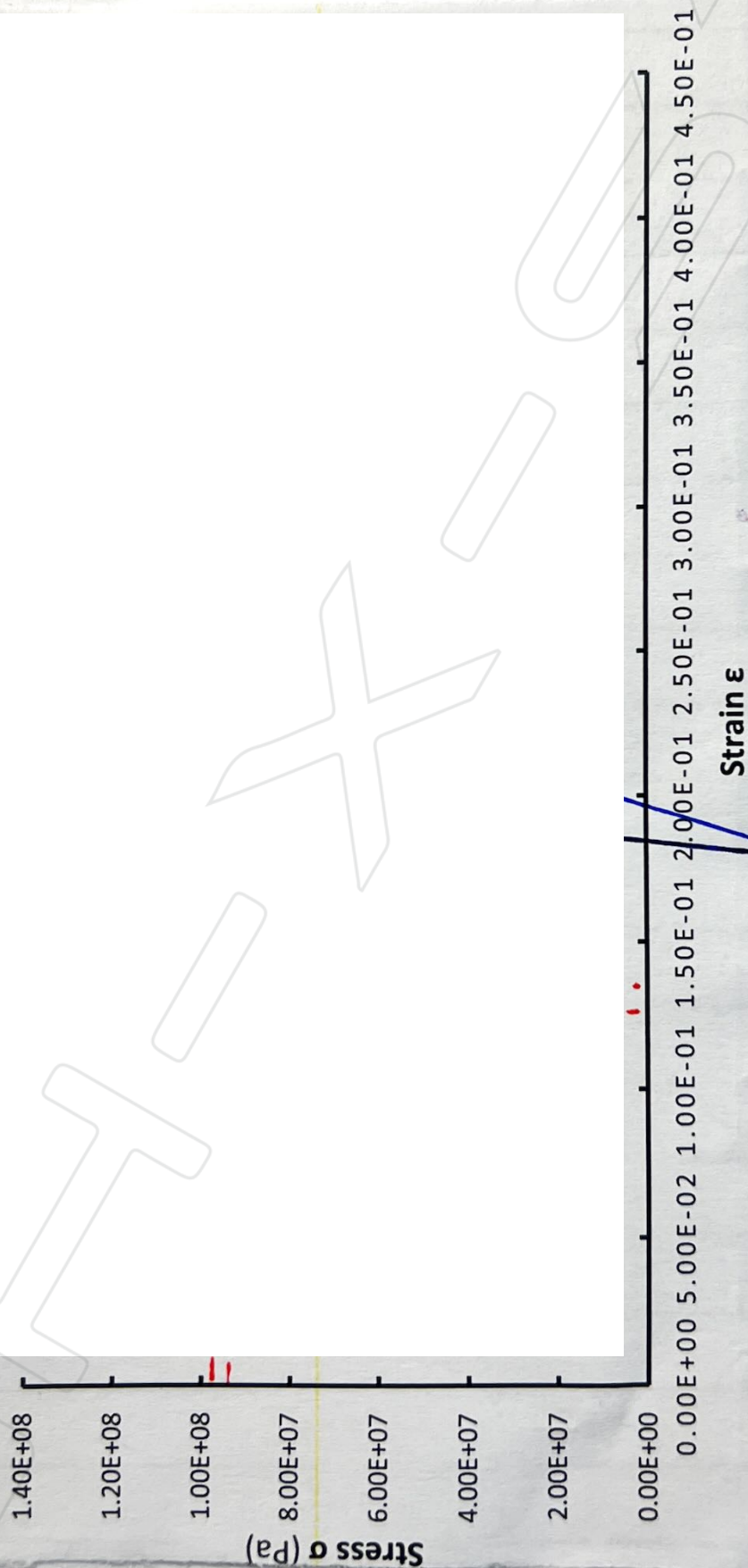
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11/30/2022 Tension and Hardness Post-Lab

Table 1 : Tensile test results

specimen	A_0 (mm ²)	l (mm)	E (MPa)	critical σ (MPa)	UTS in nominal stress (MPa)	ϵ_f in true strain	specific stiffness (mm ² /s ²)	specific strength (mm ² /s ²)
Aluminium								
Carbon fibre								
Rigid plastic								

Table 2 : Hardness Test Results

average measured hardness = 754.8523(MPa)

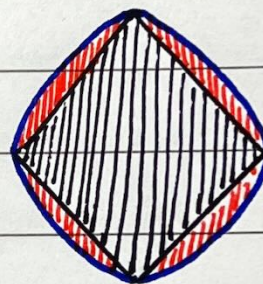
standard deviation of measured Vickers hardness = 28.5425(MPa)

hardness H^* (estimate from the tensile true stress versus strain response) = 741.7471(MPa)

Indentation conducted	d_1 (μm)	d_2 (μm)	HV (kgf/mm^2)	H (MPa)
1	160.00			
2				
3				
4				
5				

$$\bar{H} > H^*$$

inaccurated estimated hardness by larger contact area than calculated one



◇ actual area

◇ area in calculation

/// missing area by estimate